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Plant Community Biomass Shifts in Response to Mowing and Fencing in Invaded Oak Meadows with Non-Native Grasses and Abundant Ungulates

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Abstract

Many herbaceous meadows are dominated by competitive non-native grasses and subject to ungulate herbivory, ecological processes that shift the proportional biomass of plant groups in the community. Predicting the outcome of restoration is complicated because herbivory and competition can interact. We examined the relationship between herbivory by native black-tailed deer and domestic sheep and dominance of non-native grasses in Garry oak meadows, one of North America's most endangered habitat types. A 3-year factorial experiment tested the effects of mowing and fencing on plant community biomass, categorized into eight groups by geographic origin (native/non-native), growth form (annual/perennial), and plant type (forb/grass). To test if the rarity of native plant groups was related to herbivory, we estimated ungulate foraging preferences for each plant group. Mowing and fencing treatments interacted for annual and perennial non-native grasses. Dominance was shifted from non-native to native grasses only when both mowing and fencing were applied. Fencing increased the total biomass, whereas mowing had no overall effect; however, fencing alone did not affect any individual plant group. Mowing shifted dominance from grasses to forbs, although both native and non-native forbs benefited from the increased light availability. We also noted that herbivore fecal pellet densities were greatest in the spring, which coincided with the peak season of their preferred plant group, native perennial forbs. Overall, applying both mowing and fencing was the most effective restoration treatment to increase native plant groups and biomass.

Key words: **black-tailed deer, disturbance, exotic, grass, grazing, invasive, oak savanna, sheep.**

Introduction

Many plant communities are becoming increasingly homogeneous, dominated by few competitive, often non-native species (Wilsey & Polley 2006). These new states may be the result of changes to biomass altering disturbance regimes. Disturbances can be episodic and nonselective, such as burning or mowing (MacDougall & Turkington 2004, 2007), or chronic and selective, such as herbivory (Augustine et al. 1998). Unfortunately, reinstating disturbance regimes in invaded communities may not restore native plant communities given that present conditions and species composition differ from the former state (Hobbs & Huenneke 1992; Keeley et al. 2003). That many ecosystems experience concurrent changes to more than one disturbance regime further complicates restoration efforts (Cione et al. 2002). The science of predicting plant community response to disturbance is still rudimentary (Lavorel & Garnier 2002), and field experiments investigating more than one type of disturbance are rare (Hooper et al. 2005; Wisdom et al. 2006). As a consequence, there is an urgent need to improve our ability to predict the effects of different types of disturbances on plant communities for the recovery of native ecosystems and natural processes.

Communities dominated by herbaceous plants such as grasslands, prairies, and meadows are among the most invaded and endangered terrestrial habitats because they are particularly susceptible to disturbance and anthropogenic activities (Lozon & MacIsaac 1997; Lonsdale 1999; Alpert et al. 2000). Herbaceous communities are also challenging to restore because they are often fragmented, have lost native species, and have experienced altered disturbance regimes (Suding et al. 2004). Non-native grasses have come to dominate many herbaceous communities in North America. Their success has been attributed to their ability to produce clumps of living material and litter that decompose slowly, minimizing ground light levels and inhibiting germination and establishment of other species (MacDougall & Turkington 2004; MacDougall 2005). Biomass removal via burning or mowing can shift composition, structure, and function without eliminating small

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native plant populations and therefore has been a beneficial technique for native plant restoration in some herb-dominated plant communities (MacDougall 2005; MacDougall & Turkington 2007).

Deer (*Odocoileus* spp.) densities, which have increased substantially in forested regions throughout North America because of landscape change and predator eradication (Côté et al. 2004), are a threat to natural herbaceous communities (MacDougall 2008). Forests with abundant deer have become homogenized (Rooney & Waller 2003), altered in structure (Stockton et al. 2005; Navarro et al. 2006), and dominated by less palatable species (Kellogg & Bridgham 2004). Although the effects of deer have rarely been examined in herbaceous communities, domestic herbivores are known to shift plant communities to new stable states from which the native community does not recover even once the herbivore populations have been reduced or removed (Fleischner 1994; Cingolani et al. 2005).

Once widespread in North America, oak meadows and savannas have declined precipitously because of settlement and fire suppression (Nuzzo 1986; Lea 2006) and remaining patches are threatened by non-native plant species (MacDougall et al. 2004) and abundant herbivores (Gonzales & Arcese 2008). To date, restoration of Garry oak (*Quercus garryana*) meadows in western Canada (MacDougall & Turkington 2007) and in the United States (Devine et al. 2007) has involved biomass removal via burning or mowing, or fencing to reduce herbivory, but not both. Mowing tends to increase abundances of native plant species, geophytes, and annual plant species because traits that make subordinates poor competitors against non-native perennial grasses (smaller size, short growing season, and lack of underground storage capacity) favor growth and reproduction following mowing (MacDougall 2005). Species richness can also increase with mowing because of the increased availability of light at ground level (MacDougall & Turkington 2004). Fencing from herbivores can increase abundances of some native species, particularly palatable forbs (Augustine & McNaughton 1998), whereas less palatable species, such as some non-native plants, flourish under herbivory (Keane & Crawley 2002; Zou et al. 2007). Forage preferences of deer can alter plant community composition (Wiegmann & Waller 2006), and abundant deer may have facilitated the shift from native forbs to non-native grasses in Garry oak meadows (Gonzales 2008). The response of the plant community to simultaneous treatments of mowing and fencing is unknown, and interactions between the treatments may have consequences for community composition.

We expected groups of plants within the community to respond positively or negatively to the treatments according to general characteristics of those plant groups. Native annual and perennial forbs were expected to increase and non-native perennial forbs and grasses to decrease with mowing. We also predicted native annual and perennial forbs would increase and non-native perennial forbs and grasses would decrease with fencing. Non-native annual forbs and grasses tend to increase with mowing and decrease with fencing, whereas native perennial grasses decrease with mowing and increase with fencing; therefore, the effects of mowing and fencing applied to these three plant groups were expected to interact.

We also examined three other factors that could be influential to the proportional biomass of plant groups in the plant community. First, we asked if ungulates selectively consumed certain plant groups. Ungulates are known to favor certain plants such as lilies (Krausman et al. 1997), which could provide indirect benefits to less-preferred plant groups (Keane & Crawley 2002; Zou et al. 2007), altering their proportional biomass in the community. Second, given that abiotic factors influence dominance, we tested whether ground light availability, soil depth, and/or soil moisture was related to species richness. Third, given that both plants and herbivores can be seasonal, some ephemeral plant groups may experience greater herbivore pressure during the season that herbivores are present in the greatest densities. Therefore, we examined shifts in the proportional abundance of each plant group through time and also estimated herbivore activity as the season progressed.

The plots used in this experiment were part of another experiment examining the effects of herbivory and shading on experimentally added native species found commonly in Garry oak meadows (Gonzales & Arcese 2008). We tested whether experimentally added species influenced the proportional biomass of plant groups in this experiment and found no effect.

Methods

Study Site

Garry oak ecosystems extend from central California in the United States to southwestern British Columbia in Canada. Within Canada, Garry oak ecosystems are found on southeastern Vancouver Island and the Gulf Islands, an archipelago located in the Strait of Georgia. They encompass a variety of habitat types including rocky bluffs, vernal pools, deep soil savannas, and shallow soil meadows (Fuchs 2001). Our study occurred in the shallow soil meadows common to the southern Gulf Islands (Fig. 1). Crow's Nest Ecological Research Area (CNERA) on Salt Spring Island (48°46′51″N, 123°27′23″W) is 200 m above sea level with a southeast aspect. The sub-Mediterranean climate is mild and wet from November to April (mean values for Nov–April = 5°C, 128 mm) and moderate and drier from May to October (mean values for May–Oct = 13°C, 43 mm). Soils are shallow and interspersed with exposed bedrock. The 30-ha research area is comprised of two plant communities, 1.4 ha of more open Garry oak meadows with rocky outcrops and the remainder is Douglas-fir (*Pseudotsuga menziesii* (Mirbel) Franco)-dominated forest. Non-native grasses are abundant and high levels of herbivory are indicated by the modified architecture of shrubs and trees. Abundant native black-tailed deer (*Odocoileus hemionus columbianus*) and a small flock of domestic sheep are present at CNERA.

Mowing and Fencing Experiment

We crossed mowing and fencing to determine which restoration technique(s) increased native plant biomass and/or

Figure 1. Surveying vegetation in an experimental plot in a Garry oak meadow on Salt Spring Island, British Columbia, B.C., Canada. Abundant non-native perennial grasses are under the canopy of the Garry oaks (*Quercus garryana*) in the foreground. Photo by David R. Clements.

reduced non-native biomass in Garry oak meadows. A 2 × 2 factorial experiment was used given the potential for interactions between treatments. The plant community was categorized into plant groups defined by traits expected to respond to mowing and fencing: geographic origin (native/non-native), growth form (annual/perennial), and plant type (forb/grass [grasses, sedges, and rushes]) (Díaz et al. 2007). These groupings are consistent with previous community ecology studies in Garry oak ecosystems (MacDougall et al. 2006; Best & Arcese 2009). No native annual grasses were observed in these studies; therefore, we expected the plant community to consist of seven groups.

Fifty-six experimental plots were randomly selected from sites that met the following criteria: less than 1% open rock, less than 10% slope, less than 5% canopy closure, greater than 10 cm soil depth, and greater than 10 m distance to the next plot. Herbaceous plant communities can be highly variable because of local abiotic conditions (Lortie & Cushman 2007); therefore, plot selection was designed to minimize variability because of soil moisture and soil depth. Plot size, 1 m ×1 m, was selected to maximize the number of replicates meeting the plot selection criteria. Environmental variables recorded at each plot included light at ground level (LI-COR quantum sensor; LI-COR, Lincoln, NE, U.S.A.), April soil moisture (Hydrosense TDR meter; Campbell Scientific, Logan, UT, U.S.A.), and soil depth, which was measured with a 90-cm steel rod driven into the ground to bedrock. Both soil depth and soil moisture were measured in each of the four cardinal directions adjacent to the plot edges and the average was taken for each plot.

Fencing and mowing treatments were applied randomly to the balanced experiment. The treatments were initiated February 2003 and ceased August 2005. Herbivory was manipulated by fencing (F+) or leaving plots unfenced (F−). The 28 fenced plots were enclosed by 1.25-m^3 aluminum frames and large meshed fish nets to exclude ungulates without excluding light or small animals. Mowing involved cutting plots with clippers near ground level and discarding live plant material and litter (M+) or leaving plots unmowed (M−). The 28 plots were mowed in the beginning (October/November) and end (July/August) of the growing season each year so as to avoid disturbance during the growing period for native plants. Biomass collection began on 2 April 2005 and was completed on 19 July 2005. Approximately 20 person-hours per plot were required to identify and harvest the plants. Plots were harvested, in random order, by identifying to species, and storing each species by plot in a paper bag. Plants were dried in a drying oven for 48 hours at 70°C and weighed to a precision of 0.01 g.

Herbivore Forage Preferences and Herbivore Seasonal Activity

Given that the subordinance of native plant groups could be related to herbivory, we examined whether black-tailed deer and sheep selected plant groups in relation to their availability in the community using a log-linear chi-square test. To do so, we identified and counted all stems to species in sixteen 0.5 m ×0.5 m plots noting whether they were browsed and calculated forage indices for each species, as well as pooling species into plant groups. The forage index (w_i) was the proportion of the browsed stems of each species (o_i) over the proportion of the total stems of each species (μ_i):

$$w_i = o_i/\mu_i$$

Indices greater than 1 indicated selection, whereas indices below 1 suggested avoidance (Manly et al. 2002).

Following biomass collection in 2005, we observed that the proportional biomass of native perennial forbs declined as the season progressed. To infer whether declines of native perennial forbs could be exacerbated by seasonal herbivore activity, we counted fecal pellets in CNERA over five time periods from November 2005 to August 2006. Fecal pellets were counted along the same 10 transects that varied in length (mean = 65.1 ± 10.8 m) according to topography. The amount of time spent searching along the transect was kept approximately constant for each individual transect to maintain a consistent search effort (mean = 6.9 ± 0.97 minutes). A single pellet group consisted of at least 10 pellets spatially clumped together. Surveyors counted the number of pellet groups within approximately 1 m to each side of transects and then cleared the area of pellets to avoid repeated counts.

Data Analyses

We explored how mowing and fencing influenced species richness, native, non-native, and total biomass, as well as the proportional biomass of seven plant groups. Proportions, rather than absolute biomass, were specified as the response variables because plant groups within plots were not completely independent observations (Pearse & Ratti 2004). Each of the seven models included, as the main factors, mowing and fencing and their interaction as well as time as a continuous variable and the number of stems of experimentally added plants as a covariate. These plants from the other experiment were included to test whether their abundance affected the treatments in this experiment. Time was included because biomass collection spanned nearly 4 months, and the proportional biomass may have been influenced by time if phenology differed between the plant groups. Therefore, the full models included interactions of time with the main factors and the three-way interaction. No significant values were found in the higher level interactions, so we reported the simpler models.

Proc GLIMMIX (SAS Institute 2006, Cary, NC, U.S.A.) allows for the specification of binomial, Poisson, and other non-normal distributions along with link functions that relate the expectation of the dependent variable to a linear predictor (Witte et al. 2000; Wedel 2001). We used a Poisson distribution with a log link function to test whether species richness responded to mowing, fencing, or their interaction with time included as a random factor. We tested total, native, and non-native biomass with a general linear mixed model (Proc MIXED) with time included as a random factor. Residuals were assessed with Shapiro–Wilk and Bartlett's tests to determine if assumptions of normality and homoscedasticity were met. We explored transformations of the response variables with non-normal distributions (e.g., proportional biomass and richness), but not all variables could be normalized. Two extreme values were removed from native biomass to meet assumptions of normality. We used beta distributions, which added a scaling factor to accommodate overdispersion of the residuals, with logit link functions, for the analyses of proportional biomass of the seven plant groups. Goodness of fit for all generalized linear models was assessed with the ratio of chi-square and degrees of freedom (*df*) (≈ 1 indicates a good fit) (McCullagh & Nelder 1989). Significance was assessed at $\alpha \leqslant 0.05$.

Results

Mowing and Fencing Experiment

The mean (± 1 SD) biomass collected from the 56 plots was 248 g/m^2 (± 144 g/m^2). Non-native perennial grasses were the most abundant plant group (36%, 89 ± 83 g/m^2) and native annual forbs the least abundant (3%, 7 ± 10 g/m^2). Of the ten most abundant species, eight were non-native. No native annual grasses were observed. Mowed plots had greater light ($F_{[1,43]} = 7.69$, $p = 0.01$) and soil moisture availability ($F_{[1,155]} = 6.27$, $p = 0.01$) throughout the growing season compared with unmowed plots. Fencing did not alter moisture ($F_{[1,155]} = 0.22$, $p = 0.64$) or light levels ($F_{[1,43]} = 1.75$, $p = 0.19$). Species richness increased with increasing light levels ($F_{[1,54]} = 7.74$, $p = 0.01$), whereas there were no relationships between richness and soil moisture ($F_{[1,54]} = 0.59$, $p = 0.45$) and soil depth ($F_{[1,54]} = 0.15$, $p = 0.70$).

In the absence of restoration treatments, both annual and perennial non-native grasses were abundant (Fig. 2). Non-native perennial grasses had the greatest proportional biomass in all but the mowed, fenced plots, in which native perennial grasses had the greatest proportional biomass (Fig. 2). Species richness was highest in mowed plots (24 species), whereas fencing had no effect (Table 1). With all plant groups pooled together, fencing increased the total biomass by 38% relative to open plots, whereas mowing had no effect (Table 1). Total native biomass increased with both fencing (40%) and mowing (30%), whereas neither treatment affected total non-native biomass (Table 1).

As predicted, native and non-native annual forbs increased with mowing (Tables 2 & 3). Mowing doubled the proportional biomass of native annual forbs, whereas non-native annual forbs increased by 47% (Fig. 2). Mowing and fencing interacted for non-native annual grasses, as expected; however, the observed response of the proportional biomass of the other plant groups differed from our predictions (Tables 2 & 3). Mowing increased the proportional biomass of native perennial grasses by 57%, and there was an interaction between treatments for non-native annual and perennial grasses (Tables 2 & 3). For non-native annual grasses, mowed, fenced plots had greater proportional biomass by 22% relative to unmowed, fenced and by 28% relative to mowed, unfenced plots (Fig. 2). The proportional biomass of non-native perennial grasses was lowest in plots with both treatments applied (Tables 2 & 3). Untreated (control) plots and mowed, unfenced plots had, respectively, 68 and 39% more proportional biomass than mowed, fenced plots (Fig. 2). In particular, fencing without mowing increased the proportional biomass of non-native perennial grasses by 125% (Fig. 2). Moreover, unmowed, fenced plots had more proportional biomass of non-native perennial grasses than unmowed, unfenced plots by 34% and mowed, unfenced by 62% (Fig. 2). Neither treatment influenced the proportional biomass of native or non-native perennial forbs; however, the proportional biomass of native perennial forbs declined as the season progressed (Tables 2 & 3).

Herbivore Forage Preferences and Seasonal Activity

We counted a total of 9,307 stems in control plots and 12% were browsed. Native perennial forbs were browsed more often than other plant groups given their availability in the plant community (Fig. 3). Specifically, native lilies such as harvest brodiaea [*Brodiaea coronaria* (Salisb.) Engl., $w = 6.48$] and fool's onion [*Triteleia hyacinthina* (Lindl.) Greene, $w = 8.15$] were browsed more often than other species. Graminoids such as rat-tail fescue [*Vulpia myuros* (L.) C.C. Gmel., $w = 0.01$], soft brome [*Bromus hordeaceus* L., $w = 0.08$], and long-stoloned sedge [*Carex inops* L.H. Bailey, $w = 0.08$] were generally avoided given their abundance in the plant community.

Herbivore pellet deposition varied seasonally; there were twice as many pellets in March than in the other months that were surveyed (Fig. 4).

Discussion

Garry oak meadows have been degraded by non-native grasses and abundant herbivores, and we asked how two crossed

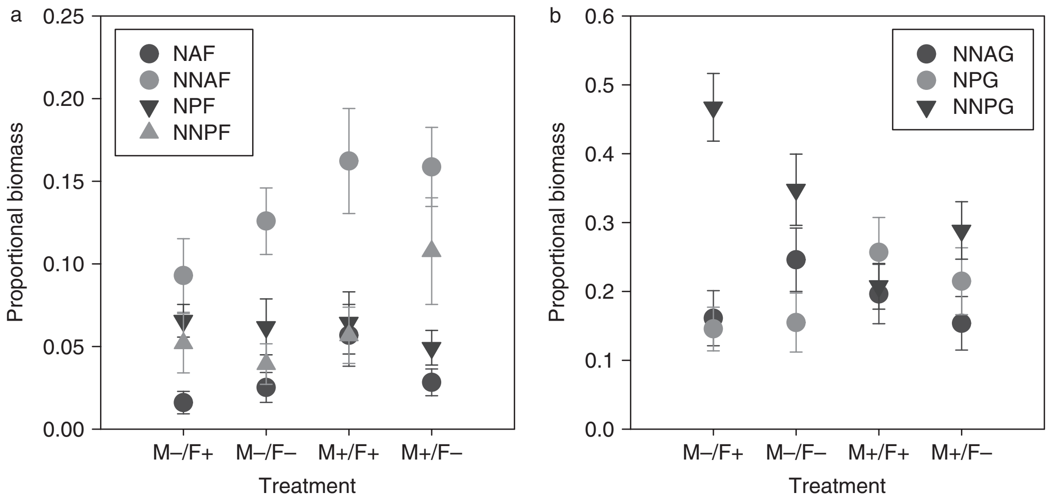

Figure 2. The proportional biomass of forbs (a) and grasses (b) in a factorial experiment consisting of unmowed (M−) and mowed (M+) treatments crossed with open (F−) and fenced (F+) treatments. Plant groups were characterized by geographic origin (native [N]/non-native [NN]), growth form (annual [A]/perennial [P]), and plant type (forb [F]/grass [G]). No native annual grasses were observed in the plant community. Error bars = +1 SE.

Table 1. The response of richness and biomass to restoration treatments (mowing [M] and fencing [F]) in a Garry oak meadow analyzed with general/generalized linear models.

Plant Community Measure	*Effect*	*df*	*F Value*	*p*	*2logL*	$\chi^2 df$
Total richness	M	1,51	5.02	**0.03**	−310.85	0.65
	F	1,51	2.6	0.11		
	M × F	1,51	0.31	0.31		
Total biomass	M	1,52	1.19	0.28		
	F	1,52	7.43	**0.01**		
	M × F	1,52	0.78	0.38		
Native biomass	M	1,49	4.68	**0.04**		
	F	1,49	7.21	**0.01**		
	M × F	1,49	0.81	0.37		
Non-native biomass	M	1,51	3.23	0.08		
	F	1,51	3.14	0.08		
	M × F	1,51	1.61	0.21		

Significance ($\alpha = 0.05$) is shown in bold type. The −2LogL and χ^2/df values are given for the generalized linear model to assess fit.

restoration treatments, mowing and fencing, affected the plant community. In particular, we tested the efficacy of mowing and fencing on shifting the proportional biomass from non-native to native plant groups. Overall, mowing and fencing increased the proportional biomass of native plants, but each technique had differing strengths and effects, which interacted for some plant groups, complicating the restoration prescription.

The interaction of mowing and fencing prevents a straightforward interpretation of the results. Nevertheless, we can infer that non-native perennial grasses have a suppressive effect on other plant groups and that the absence of disturbance facilitates their dominance on the basis of three results. First, mowing increased native and non-native annual forbs, consistent with the hypothesis that the suppression of nonselective, episodic disturbances such as mowing or fire facilitated their decline (MacDougall & Turkington 2005, 2007). Second, native perennial grasses increased with mowing and also became the most abundant plant group in the mowing and fencing treatment, whereas non-native perennial grasses were the most abundant plant group in both of the unmowed treatments. Third, other than the unmowed, unfenced treatment with which it did not differ, the greatest proportional biomass for non-native perennial grasses was in the treatment with the least disturbance, the unmowed, fenced plot.

Cumulatively, there were larger differences in total biomass between fencing treatments than between mowing treatments. These differences could be because herbivory is a chronic disturbance whereas mowing is episodic (Wisdom et al. 2006). Fencing benefited native biomass when native plant groups were pooled and native perennial forbs were preferentially browsed; however, fencing did not benefit the proportional biomass of native forbs as expected. Two aspects of our methodology may explain this apparent discrepancy. First, we selected biomass as the response variable to directly

Table 2. The response of the proportional biomass to restoration treatments (mowing [M] and fencing [F]) in a Garry oak meadow analyzed with generalized linear models.

Plant Group	*Effect*	*df*	*F Value*	*p*	*−2logL*	*$\chi^2 df$*	*Expected*	*Observed*
Native annual forbs	M	1,43	5.49	**0.02**	−278.01	1.4	+	+
	F	1,43	0.02	0.88			+	0
	M × F	1,43	2.08	0.16			No	No
	Time	1,43	2.99	0.09				
	Exp. Plants		0.42	0.52				
Non-native annual forbs	M	1,43	3.98	**0.05**	−115.29	1.15	+	+
	F	1,43	1.72	0.2			–	0
	M × F	1,43	1.17	0.29			Yes	No
	Time	1,43	0.17	0.75				
	Exp. Plants		0.03	0.86				
Native perennial forbs	M	1,48	0.1	0.75	−191.13	1.08	+	0
	F	1,48	2.13	0.15			+	0
	M × F	1,48	0.23	0.63			No	No
	Time	1,48	10.58	<**0.01**				
	Exp. Plants		0.42	0.52				
Non-native perennial forbs	M	1,40	0.78	0.38	−180.76	0.87	–	0
	F	1,40	0.04	0.84			–	0
	M × F	1,40	0.01	0.91			No	No
	Time	1,40	0.53	0.47				
	Exp. Plants		0.450	0.51				
Non-native annual grasses	M	1,43	1.2	0.28	−94.57	0.99	+	0
	F	1,43	1.2	0.28			–	0
	M × F*	1,43	5.45	**0.02**			Yes	Yes
	Time	1,43	1.49	0.23				
	Exp. Plants		3.87	0.06				
Native perennial grasses	M	1,42	4.59	**0.04**	−68.18	1.17	–	+
	F	1,42	0.01	0.9			+	0
	M × F	1,42	0.02	0.89			Yes	No
	Time	1,42	0.9	0.35				
	Exp. Plants		2.93	0.09				
Non-native perennial grasses	M	1,43	19.31	<0.0001	−57.02	1.21	–	0
	F	1,43	0.14	0.71			–	0
	M × F*	1,43	8.48	**0.01**			No	Yes
	Time	1,43	3.17	0.08				
	Exp. Plants		1.25	0.27				

Time was included to test whether the proportional biomass changed during the four months that it took to collect the biomass from the plots. Significance ($\alpha = 0.05$) is shown in bold type. The −2LogL and χ^2/df values are given to assess fit. The seven plant groups were characterized by geographic origin (native/non-native), growth form (annual/perennial), and plant type (forb/grass). No native annual grasses were observed in the plant community. Expected and observed responses for each plant group to mowing and fencing are given as "+" = increase, "−" = decrease, and "0" = no difference. For mowing × fencing, "Yes" = interaction and "No" = no interaction.
*See Table 3 for comparisons between crossed treatments.

Table 3. Comparisons between crossed treatments (mowing [M] and fencing [F]) for plant groups with significant interactions.

Plant Groups	*Plot 1*	*Plot 2*	*t test*	*p*
Non-native annual grasses	M−/F−	M−/F+	$t = 0.56$	0.58
	M−/F−	M+/F−	$t = 0.95$	0.35
	M−/F−	M+/F+	$t = -1.47$	0.15
	M−/F+	M+/F−	$t = 0.24$	0.81
	M−/F+	M+/F+	$t = -2.30$	**0.03**
	M+/F−	M+/F+	$t = -2.16$	**0.04**
Non-native perennial grasses	M−/F−	M−/F+	$t = -1.68$	**0.1**
	M−/F−	M+/F−	$t = 1.17$	0.25
	M−/F−	M+/F+	$t = 2.87$	**0.01**
	M−/F+	M+/F−	$t = 2.80$	**0.01**
	M−/F+	M+/F+	$t = 4.86$	**< 0.0001**
	M+/F−	M+/F+	$t = 2.02$	**0.05**

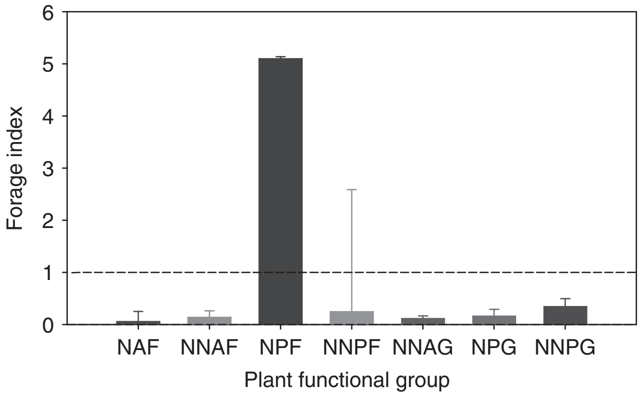

Figure 3. Forage selectivity indices for plant groups characterized by geographic origin (native [N]/non-native [NN]), growth form (annual [A]/perennial [P]), and plant type (forb [F]/grass [G]). No native annual grasses were found in the community. Values greater than 1 indicate selection in greater proportion than the availability of the plant group in the community. Error bars = +1 SE.

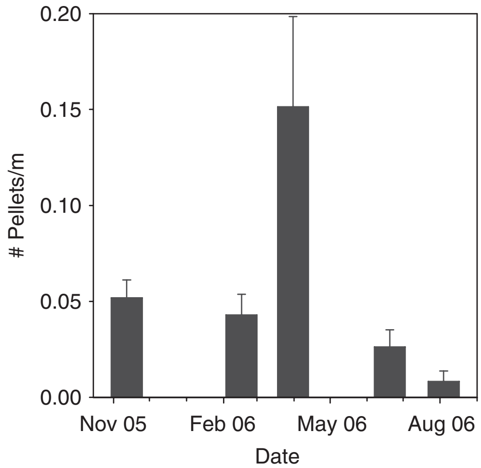

Figure 4. Seasonal herbivore activity measured as the number of pellets along 2-m belt transects ($n = 10$) in a Garry oak meadow. Error bars = +1 SE.

measure the effect of biomass removing disturbances on the community. However, biomass gives lower estimates for forbs relative to other measures of composition, such as cover, because forbs have higher water content than grasses and produce less dry weight biomass on average. Forbs also produce little persistent litter and have a shorter canopy than grasses. Therefore, biomass may not be directly linked to function in oak meadows (MacDougall 2005), and restoration strategies should consider attributes other than biomass. Second, herbivore activity was greatest in March, when many native forbs are at peak production. The biomass of native perennial forbs declined with time and our data likely captured the response of grasses to the treatments, but may have been too late in the season to capture variability among native forbs. It is possible that herbivores exerted greater pressure on native perennial forbs before April, when we began collecting biomass.

The long-term response of Garry oak plant communities to mowing and fencing treatments is unknown because as the proportion of plant groups change, local functioning is also likely to change. For example, a shorter forb canopy and increased decomposition of plant material releases nutrients at a faster rate than grasses (Garibaldi et al. 2007). Increasing establishment rates (Clark et al. 2005) of presently subordinate species could result in a more stable community where the frequency or intensity of mowing could be reduced, particularly if species diversity increases with increasing light availability (Hooper et al. 2005). Survival and persistence of experimentally added grassland forbs was not improved by 2 years of successive, frequent mowing relative to 1 year (Williams et al. 2007). It is also possible that there are limits on the ability of communities to revert to native dominance that depends on present species composition. For example, non-native plants in disturbed sites can form stable communities that are resistant to reinvasion by natives and resilient to losses of individual non-native species (Kulmatiski 2006). Community structure and composition may change with time, but non-native early colonizers may be replaced by non-native late colonizers rather than native species (MacDougall & Turkington 2005; Kulmatiski 2006). Native species are recruitment limited in invaded oak meadows, and their subordinance may be due to their rarity rather than due to their competitive inferiority (Seabloom et al. 2003a; MacDougall & Turkington 2006). Therefore, augmentation of native species (Seabloom et al. 2003b; MacDougall & Turkington 2005) coupled with mowing and fencing may provide the most rapid shifts from non-native to native species dominance. Ultimately, long-term studies that manipulate treatment levels such as the intensity and frequency of mowing and fencing, along with native species additions, offer the best chance to advance the field and increase the efficacy of restoration.

Implications for Practice

- Many herbaceous meadows are dominated by non-native grasses and subject to abundant herbivores. Mowing to reduce non-native grasses and fencing to reduce herbivory can increase native biomass, but the effects on the relative proportion of different plant groups in the community are complex because treatments interact.
- When two ecological stressors are acting on a community, it may be necessary to address both with restoration treatments to achieve the desired outcome. For example, although herbivory is a stressor for native perennial forbs, fencing without mowing can increase the proportional biomass of non-native perennial grasses. Dominance can be shifted from non-native to native grasses when both fencing and mowing are applied.
- Monitoring at the plant community level is important to understand how restoration treatments affect dominance.
- Mowing before and after the native plant growing season can maintain increased ground light levels throughout the growing season without disturbing native plants.
- Herbivore activity can be seasonal and should be monitored along with the response of the plant community to fencing and mowing.

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